

$^{34}\text{S}(\text{n},\text{n}),(\text{n},\gamma):\text{resonances}$ **1984Ca14**

1984Ca14: 30-1500-keV neutrons were produced from the Oak Ridge Electron Linear Accelerator (ORELA). The target was 94.33%-enriched ^{34}S with a 5.54% impurity of ^{32}S . Neutrons were detected using a 2-cm-thick and 7.5-cm in diameter NE110 proton-recoil detector. Prompt γ rays from neutron capture were detected using two nonhydrogenous fluorocarbon scintillators (NE-226). Measured $\sigma(E_n)$. Deduced 81 resonances, J, π , L, widths and spectroscopic factors by comparing the R-matrix predictions of the cross section with the data.

2018MuZY: Atlas of Neutron Resonances. Evaluation of neutron resonance energies, spins, width parameters, and resonance strengths for nuclei of Z=1-60.

 ^{35}S Levels

J^π and L are from **2018MuZY**. Resonance parameters E_n , $g\Gamma_n$, Γ_γ , and $g\Gamma_n\Gamma_\gamma/\Gamma$ are from **1984Ca14** due to significant discrepancies, including orders of magnitude inconsistencies, between the **1984Ca14** original data and **2018MuZY** evaluation. $g\Gamma_n=(2J+1)\Gamma_n/2$.

$E(\text{level})^\dagger$	J^π	$g\Gamma_n\Gamma_\gamma/\Gamma$	L	$E_n(\text{lab})$ (keV)	Comments
6985.54?	1/2		0	-0.305	E(level): fictitious level from analysis of resonance spectrum below the S(n) threshold. $\Gamma_\gamma=0.38$ eV.
7018.89 4	1/2	0.025 eV 1	[2]	34.03 1	
7072.66 5		0.14 eV 4		89.40 3	
7074.33 6		0.084 eV 3		91.12 4	
7097.72 30		0.009 eV 3	[2]	115.2 3	
7099.85 20		0.033 eV 5	[2]	117.4 2	$g\Gamma_n<1$ eV.
7100.73 4	3/2		[1]	118.30 1	$g\Gamma_n=372$ eV 3; $\Gamma_\gamma=0.70$ eV 4.
7143.27 11		0.42 eV 1	[2]	162.1 1	$g\Gamma_n<1$ eV.
7210.42 5			[1]	231.25 2	$g\Gamma_n=62$ eV 4; $\Gamma_\gamma=0.30$ eV 2.
7218.0 4		0.33 eV 2	[2]	239.0 4	$g\Gamma_n<4$ eV.
7234.0 5		0.19 eV 1	[2]	255.5 5	$g\Gamma_n<4$ eV.
7239.8 6		0.06 eV 2	[2]	261.5 6	$g\Gamma_n<2$ eV.
7253.2 6		0.35 eV 2	[2]	275.3 6	$g\Gamma_n<1$ eV.
7275.93 5	1/2		0	298.70 3	$g\Gamma_n=9600$ eV 150; $\Gamma_\gamma=2.82$ eV 67.
7279.13?		0.10 eV 3		302	
7289.8 8		0.31 eV 3	[2]	313.0 8	$g\Gamma_n<6$ eV.
7294.19 6	3/2		1	317.51 5	$g\Gamma_n=2500$ eV 25; $\Gamma_\gamma=0.23$ eV 8.
7331.00 5	1/2		0	355.41 2	$g\Gamma_n=4810$ eV 80; $\Gamma_\gamma=0.30$ eV 30.
7338.18 20			[1]	362.8 2	$g\Gamma_n=230$ eV 10; $\Gamma_\gamma=1.06$ eV 7.
7343.91 20		0.72 eV 3	[2]	368.7 2	$g\Gamma_n<2$ eV.
7347.2 5		0.44 eV 6	[2]	372.1 5	$g\Gamma_n<2$ eV.
7347.9 5		0.30 eV 5	[2]	372.8 5	$g\Gamma_n<2$ eV.
7353.9 5		0.44 eV 4	[2]	379.0 5	$g\Gamma_n<10$ eV.
7357.6 5		0.16 eV 4	[2]	382.8 5	$g\Gamma_n<6$ eV.
7370.56 6	1/2		1	396.14 4	$g\Gamma_n=6430$ eV 150; $\Gamma_\gamma=3.08$ eV 20.
7371.97?		0.12 eV		397.6	
7396.0 6		0.44 eV 33	[2]	422.3 6	
7404.6 6		0.21 eV 5	[2]	431.2 6	$g\Gamma_n<1$ eV.
7408.68 11	3/2		1	435.4 1	$g\Gamma_n=1570$ eV 40; $\Gamma_\gamma=0.91$ eV 8.
7411.55 6			[2]	438.35 5	$g\Gamma_n=49$ eV 5.
7416.50 4			[2]	443.45 1	$g\Gamma_n=80$ eV 8; $\Gamma_\gamma=0.51$ eV 5.
7429.1 6		0.12 eV 3	[2]	456.4 6	$g\Gamma_n<6$ eV; $\Gamma_\gamma=0.12$ eV 3.
7433.56 5			[2]	461.01 2	$g\Gamma_n=50$ eV 5; $\Gamma_\gamma=0.035$ eV 20.
7436.4 4			[1]	463.9 4	$g\Gamma_n=296$ eV 30; $\Gamma_\gamma=0.28$ eV 3.
7442.14 5	1/2		0	469.85 2	$g\Gamma_n=1050$ eV 35; $\Gamma_\gamma=0.37$ eV 8.
7462.2 6		0.33 eV 4	[2]	490.5 6	$g\Gamma_n=50$ eV 10.
7481.15 6	1/2		1	510.02 4	$g\Gamma_n=375$ eV 25; $\Gamma_\gamma=2.1$ eV 1.
7494.5 6		0.17 eV 4	[2]	523.8 6	$g\Gamma_n<4$ eV.

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{n,n}),(\text{n},\gamma)$:resonances **1984Ca14** (continued) ^{35}S Levels (continued)

E(level) [†]	J ^π	$g\Gamma_n\Gamma_\gamma/\Gamma$	L	$E_n(\text{lab})$ (keV)	Comments
7542.3 6		0.21 eV 3	[2]	573.0 6	$g\Gamma_n < 1$ eV.
7604.0 6		0.90 eV 8	[2]	636.5 6	$g\Gamma_n < 10$ eV.
7609.26 5	3/2		1	641.93 3	$g\Gamma_n = 2520$ eV 60; $\Gamma_\gamma = 0.51$ eV 6.
7648.9 6		0.21 eV 3	[2]	682.7 6	$g\Gamma_n < 30$ eV.
7655.7 6		0.80 eV 5	[2]	689.7 6	$g\Gamma_n < 10$ eV.
7663.91 11			[2]	698.2 1	$g\Gamma_n = 305$ eV 50; $\Gamma_\gamma = 0.80$ eV 8.
7678.3 7		0.45 eV 7	[2]	713.0 7	$g\Gamma_n < 4$ eV.
7731.0 7		0.31 eV 13	[2]	767.3 7	$g\Gamma_n < 4$ eV.
7749.7 4	(3/2)		[1]	786.5 4	$g\Gamma_n = 18$ eV 6; $\Gamma_\gamma = 2.65$ eV 18.
7761.46 7	3/2		1	798.65 6	$g\Gamma_n = 25350$ eV 300; $\Gamma_\gamma = 1.52$ eV 50.
7776.17 11	1/2		1	813.8 1	$g\Gamma_n = 860$ eV 60; $\Gamma_\gamma = 0.19$ eV 13.
7797.99 9	1/2		0	836.27 8	$g\Gamma_n = 3600$ eV 160; $\Gamma_\gamma = 0.39$ eV 18.
7812.24 7	3/2		2	850.94 6	$g\Gamma_n = 2650$ eV 110; $\Gamma_\gamma = 0.24$ eV 19.
7853.25 7	3/2		1	893.17 6	$g\Gamma_n = 4950$ eV 160; $\Gamma_\gamma = 0.38$ eV 16.
7861.8 12		1.13 eV 19	[2]	902.0 12	$g\Gamma_n < 3$ eV.
7880.8 12		0.67 eV 17	[2]	921.5 12	$g\Gamma_n < 20$ eV.
7889.0 12		0.57 eV 25	[2]	930.0 12	$g\Gamma_n < 20$ eV.
7894.63 7	3/2		1	935.78 6	$g\Gamma_n = 4100$ eV 160; $\Gamma_\gamma = 3.48$ eV 23.
7899.70 30			[2]	941.0 3	$g\Gamma_n = 176$ eV 16; $\Gamma_\gamma = 0.5$ eV 3.
7934.1 15		0.5 eV 3	[2]	976.4 15	$g\Gamma_n < 2$ eV.
7939.5 15		0.89 eV 4	[2]	982.0 15	$g\Gamma_n < 8$ eV.
7954.96 11	3/2		1	997.9 1	$g\Gamma_n = 930$ eV 80; $\Gamma_\gamma = 3.51$ eV 19.
7974.19 11	3/2		1	1017.7 1	$g\Gamma_n = 4020$ eV 180; $\Gamma_\gamma = 1.22$ eV 17.
8019.5 2			[2]	1064.4 2	$g\Gamma_n = 1032$ eV 90.
8041.20 20			[2]	1086.7 2	$g\Gamma_n = 2650$ eV 160.
8076.46 20			[2]	1123.0 2	$g\Gamma_n = 5480$ eV 388.
8077.91 30			[2]	1124.5 3	$g\Gamma_n = 4730$ eV 320.
8093.0 5	1/2		1	1140.0 5	$g\Gamma_n = 34880$ eV 830.
8143.66 20			[2]	1192.2 2	$g\Gamma_n = 5510$ eV 290.
8187.17 30	1/2		1	1237.0 3	$g\Gamma_n = 2945$ eV 230.
8211.25 30			[1]	1261.8 3	$g\Gamma_n = 975$ eV 86.
8224.2 4	1/2		1	1275.1 4	$g\Gamma_n = 2170$ eV 180.
8228.3 4			[2]	1279.4 4	$g\Gamma_n = 1140$ eV 100.
8243.98 30			[2]	1295.5 3	$g\Gamma_n = 1980$ eV 150.
8256.31 20	3/2		1	1308.2 2	$g\Gamma_n = 12475$ eV 770.
8262.4 4	5/2		2	1314.5 4	$g\Gamma_n = 6240$ eV 500.
8273.3 4	1/2		1	1325.7 4	$g\Gamma_n = 1635$ eV 145.
8298.27 30			[2]	1351.4 3	$g\Gamma_n = 14470$ eV 650.
8333.8 4	3/2		1	1388.0 4	$g\Gamma_n = 28845$ eV 1057.
8335.9 4			[2]	1390.1 4	$g\Gamma_n = 1955$ eV 170.
8391.3 4	3/2		1	1447.2 4	$g\Gamma_n = 27100$ eV 1390.
8393.64 30			[2]	1449.6 3	$g\Gamma_n = 3025$ eV 260.
8406.55 30			[2]	1462.9 3	$g\Gamma_n = 3885$ eV 345.
8418.8 4			[2]	1475.5 4	$g\Gamma_n = 3795$ eV 340.

[†] From $E_{c.m.} + S(n)$ where $S(n) = 6985.84$ 4 (2021Wa16) and $E_{c.m.} = E_n(\text{lab}) \times m(^{34}\text{S}) / [m_n + m(^{34}\text{S})]$.